

Recursive Estimation	
Fabio Bühler, fabuehle@ethz.ch Version: 4. August 2024	
1 Probability Theory	
1.1 Random Variables	
1.1.1 Discrete Random Variables	
DRV – Discrete Random Variable defined by $\mathcal{X}, p_x: \mathcal{X} \rightarrow \mathbb{R}$	
$\mathcal{X} \subset \mathbb{Z}$ of all possible outcomes $p_x(\bar{x}) \geq 0$ and $\sum_{\bar{x} \in \mathcal{X}} p_x(\bar{x}) = 1$ or $\sum_{\bar{x} \in \mathcal{X}} p_{x y}(\bar{x} \bar{y}) = 1$	
Margin. / Sum Rule	$p(\bar{x}) = \sum_{\bar{y} \in \mathcal{Y}} p_{xy}(\bar{x}, \bar{y})$
Cond. / Product Rule	$p_{x y}(\bar{x} \bar{y}) := \frac{p_{xy}(\bar{x}, \bar{y})}{p_y(\bar{y})}$
Total Prob. Theorem	$p_x(\bar{x}) := \sum_{\bar{y} \in \mathcal{Y}} p_{x y}(\bar{x} \bar{y})p_y(\bar{y})$
1.1.2 Continuous Random Variables	
CRV – Continuous Random Variable: $\text{DRV} \sum \rightsquigarrow \text{CRV} \int$	
- Set $\mathcal{X} \subset \mathbb{R}$ section of real line - PDF p_x satisfies $p_x(\bar{x}) \geq 0$ and $\int_{\mathcal{X}} p_x(\bar{x})d\bar{x} = 1$	
Caution: $Pr(x = \bar{x}) = 0, \quad Pr(x \in (a, b)) = \int_a^b p(\bar{x})d\bar{x}$	
1.1.3 Conditional PDF	
Conditional PDF	
Margin. / Sum Rule	$p(\bar{x} \bar{z}) = \sum_{\bar{y} \in \mathcal{Y}} p_{xy z}(\bar{x}, \bar{y} \bar{z})$
Cond. / Product Rule	$p_{x yz}(\bar{x} \bar{y}, \bar{z}) := \frac{p_{xy z}(\bar{x}, \bar{y} \bar{z})}{p_{y z}(\bar{y} \bar{z})}$
Independence	
$p(x y) = p(x) \Leftrightarrow p(y x) = p(y) \Leftrightarrow p(x, y) = p(x)p(y)$ $p(x, y, z) = p(x, y z)p(z) \rightarrow x, y$ indep. $p(x, y z) = p(z x, y)$	
Conditional Independence	
The knowledge of z makes x and y independent: $p(x y, z) = p(x z) \Leftrightarrow p(y z) = p(x z)p(y z)$	
Caution!!! in general we still have: $p(x, y) \neq p(x)p(y)$ Caution!!! Independence $\nRightarrow$ Conditional Independence	
1.2 Expectation and Variance	
1.2.1 Expectation	
Definition: <i>Integral for CRV</i> $\mathbb{E}_x[x] = \sum_{\bar{x} \in \mathcal{X}} \bar{x}p_x(\bar{x})$	
Linearity: $\mathbb{E}_{xy}[a + bx + cy] = a + b\mathbb{E}_x[x] + c\mathbb{E}_y[y]$ Multi Variable: $\mathbb{E}_{xy}[g(x, y)] = \sum_{\bar{y}} \sum_{\bar{x}} \bar{x}g(\bar{x}, \bar{y})p_{xy}(\bar{x}, \bar{y})$ Independence of x, y $\Rightarrow \mathbb{E}_{xy}[xy] = \mathbb{E}_x[x]\mathbb{E}_y[y]$	
Law of Unconscious Statistician for $y = g(x)$	
$\mathbb{E}_y[y] = \sum_{\bar{y} \in \mathcal{Y}} \bar{y}p_y(\bar{y}) = \sum_{\bar{x} \in \mathcal{X}} g(\bar{x})p_x(\bar{x})$	
1.2.2 Variance (generally a matrix)	
$\text{Var}_x[x] = \mathbb{E}_x \left[(x - \mathbb{E}_x[x])(x - \mathbb{E}_x[x])^\top \right] \stackrel{1D}{=} \mathbb{E}_x \left[x^2 \right] - \mathbb{E}_x[x]^2$	
Linearity	
$\text{Var}_x[a + bX + cY] = b^2 \text{Var}_X[X] + c^2 \text{Var}_Y[Y] + 2bc \text{Cov}[X, Y]$	
Covariance: if independent $\text{Cov}(X, Y) = 0$	
$\text{Cov}[X, Y] = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$	
1.3 Distribution Sampling	
CDF – Cumulative Distribution Function	
$\hat{F}_x(\bar{x}) := \sum_{i=-\infty}^{\bar{x}} \hat{p}_x(i) = Pr(x \leq \bar{x}) = \int_{-\infty}^{\bar{x}} p_x(\hat{x})d\hat{x}$	
A1 – 1 DRV $\bar{u}$ sample of $u \sim \mathcal{U}$, solve for $\bar{x}$ such that: $\hat{F}_x(\bar{x} - 1) < \bar{u} \leq \hat{F}_x(\bar{x})$	
A2 – Multiple DRV or CRV Works for finite & infinite # of elements	

- Decompose $\hat{p}(\bar{x}, \bar{y}) = \hat{p}_{x y}(\bar{x} \bar{y})\hat{p}_y(\bar{y})$ - Apply A1 to get sample $\bar{y}$ via $\hat{p}_y(\bar{y})$ - With $\bar{y}$ fixed, apply A1 to get sample $\bar{x}$ via $\hat{p}_{x y}(\bar{x} \bar{y})$	A3 – 1 CRV $\hat{p}_x$: pw-conti., bounded. $u \sim \mathcal{U}(0, 1)$ and solve: $\bar{u} = F_x(\bar{x}) \rightarrow \bar{x} = F_x^{-1}(\bar{u})$
1.4 Change of Variables	
DRV: $x = g(y)$	$p_x(\bar{x}) = \sum_{\bar{y} \in \mathcal{Y}: g(\bar{y}) = \bar{x}} p_y(\bar{y})$
CRV general case $y = g(x), \rightarrow x = g^{-1}(y)$:	$p_y(\bar{y}) = \left \frac{dg}{dx}(\bar{x}) \right , \quad \bar{x} = g^{-1}(y)$
CRV – Multivar. & Indep. Case: $z = x + y$	$p_z(\bar{z}) = \sum_{\bar{x} \in \mathcal{X}} p_x(\bar{x})p_y(\bar{z} - \bar{x})$
CRV – Multivar. & Cond. Case: $z = g(w, x), w = h(z, x)$	Assumptions: - $g(w, x), p_w$ cont. diff'bar - $g(w, x)$ strictly monotonic - For scalars $\det(x)$ is just x
1.5 Miscellaneous	
Indep. multiplication: $z = xy \rightarrow p(z) = \sum_Y p_x(\frac{z}{y})p_y(y)$	
Uniform Distribution	
$p(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{else} \end{cases}$	$\mathbb{E}\{x\} = \frac{a+b}{2}$ $\text{Var}\{x\} = \frac{(b-a)^2}{12}$
Triangular Distribution	
$p(x) = \begin{cases} \frac{2(x-a)}{(b-a)(c-a)} & a \leq x < c \\ \frac{2(b-x)}{(b-a)(b-c)} & c \leq x \leq b \\ 0 & \text{else} \end{cases}$	
$F(x) = \begin{cases} 0 & x < a \\ \frac{(x-a)^2}{(b-a)(c-a)} & a \leq x \leq c \\ 1 - \frac{(b-x)^2}{(b-a)(b-c)} & c \leq x \leq b \\ 1 & x > b \end{cases}$	
$\mathbb{E}\{x\} = \frac{a+b+c}{3}, \quad \text{Var}\{x\} = \frac{a^2+b^2+c^2-ab-ac-bc}{18}$	
Gaussian Distribution $\mathbf{x} = (x_1, \dots, x_D) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	
$p_{\mathbf{x}}(\bar{\mathbf{x}}) = \frac{1}{(2\pi)^{D/2} \det(\boldsymbol{\Sigma})^{1/2}} \exp \left(-\frac{(\bar{\mathbf{x}} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\bar{\mathbf{x}} - \boldsymbol{\mu})}{2} \right)$ $\rightsquigarrow$ fully characterized by mean $\boldsymbol{\mu} \in \mathbb{R}^D$ and Variance $\boldsymbol{\Sigma} \in \mathbb{R}^{D \times D}$ (Scalar: $D = 1, \sigma^2 = \det(\boldsymbol{\Sigma}) \rightarrow \det(\boldsymbol{\Sigma})^{1/2} = \sigma$)	
$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(x - \mu)^2}{2\sigma^2} \right)$	$\left. \begin{array}{l} \mathbb{E}\{x\} = \mu \\ \text{Var}\{x\} = \sigma^2 \end{array} \right $
Affine transformation of Gaussian: $x \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	
$\mathbf{y} = M\mathbf{x} + \mathbf{b} \rightarrow \mathbf{y} \sim \mathcal{N}(M\boldsymbol{\mu} + \mathbf{b}, M\boldsymbol{\Sigma}M^T)$	
Linear combination of jointly GRV: $\mathbf{x}, \mathbf{y}$	
$\mathbf{z} = M_x\mathbf{x} + M_y\mathbf{y} \Rightarrow \mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}_z, \boldsymbol{\Sigma}_z)$	
$\boldsymbol{\mu}_z = M_x\boldsymbol{\mu}_x + M_y\boldsymbol{\mu}_y$	
$\boldsymbol{\Sigma}_z = M_x\boldsymbol{\Sigma}_xM_x^T + M_x\boldsymbol{\Sigma}_{xy}M_y + M_y\boldsymbol{\Sigma}_{xy}^T M_x + M_y\boldsymbol{\Sigma}_yM_y^T$	
(Red symbols are zero if $\mathbf{x}, \mathbf{y}$ are independent)	
Diagonal Variance $\rightsquigarrow y_i$ mut.indep. iff $\boldsymbol{\Sigma}$ diagonal	
$p(y) = \prod_{i=1}^D \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp \left(-\frac{(y_i - \mu_i)^2}{2\sigma_i^2} \right)$	

2 Bayesian Tracking
2.1 Bayes Theorem
Definition: (Applies to discrete and continuous random variables)
$p(x y) = \frac{p(y x)p(x)}{p(y)}$
x =system state, $p(x)$ =prior belief of state, z =observations related to state $p(z x)$ =observation likelihood, $p(x z)$ =posterior belief of state $p(z)$ =normalization constant $= \sum_x p(z x)p(x)$ For multiple observations:
$\underbrace{p(x z_1, \dots, z_N)}_{\text{posterior}} = \underbrace{\sum_{x \in \mathcal{X}} p(x)}_{\text{normalization}} \underbrace{\prod_i \overbrace{p(z_i x)}^{\text{observation likelihood}}}_{\text{prior}}$
2.2 Bayesian Tracking
Model
Objective: Calculate pdf of state $p(x(k) z(1:k))$
$x(k) = q_{k-1}(x(k-1), v(k-1)) \in \mathcal{X}$
$z(k) = h_k(x(k), w(k))$
Prior Update - State Prediction
$p(x(k) z(1:k-1)) = \sum_{x(k-1)} \underbrace{p(x(k) x(k-1))}_{\text{process model}} \underbrace{p(x(k-1) z(1:k-1))}_{\text{previous iteration}}$
Measurement Update
$p(\bar{x}(k) \bar{z}(1:k)) = \underbrace{\frac{p_z(\bar{z}(k) \bar{x}(k))}{\sum_{i \in \mathcal{X}} p(\bar{z}(k) i)} \overbrace{p(\bar{x}(k) \bar{z}(1:k-1))}^{\text{prior}}}_{\text{normalization}}$
Computer Implementation
1. Enumerate state space $\mathcal{X} = \{0, 1, \dots, N-1\}$ 2. Posterior define $\mathbf{a}_{k k}^i := p_{x(k) z(1:k)}(i \bar{z}(1:k))$ 3. Prior define $\mathbf{a}_{k k-1}^i := p_{x(k) z(1:k-1)}(i \bar{z}(1:k-1))$
Initialization $\mathbf{a}_{0 0} = p_{x(0)}(i) \quad i = 0, \dots, N-1$
$\left. \begin{array}{l} \mathbf{a}_{k k-1}^i = \sum_{j=0}^{N-1} p_{x(k) x(k-1)}(i j) \mathbf{a}_{k-1 k-1}^j \\ \mathbf{a}_{k k}^i = \frac{p_z(k) x(k)}{\sum_{j=0}^{N-1} p_z(k) x(k)} \frac{(\bar{z}(k) i) \mathbf{a}_{k k-1}^i}{(\bar{z}(k) j) \mathbf{a}_{k k-1}^j} \end{array} \right\} \text{Iterate } k > 0$
3 Extracting Estimates
3.1 General Measurement
$\underbrace{\begin{bmatrix} z_1 \\ \vdots \\ z_m \end{bmatrix}}_{z \in \mathbb{R}^m} = \underbrace{\begin{bmatrix} h_{11} & h_{1n} \\ \vdots & \vdots \\ h_{m1} & h_{mn} \end{bmatrix}}_{H \in \mathbb{R}^{m \times n}} \cdot \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{x \in \mathbb{R}^n} + \underbrace{\begin{bmatrix} w_1 \\ \vdots \\ w_m \end{bmatrix}}_{w \in \mathbb{R}^m}$
m Measurements, n States, $w_i \sim \mathcal{N}(0, R)$ – Zero-mean additive gaussian, $x, w_i \forall i$ assumed mutually independent
3.2 General Measurement
$\underbrace{\begin{bmatrix} z_1 \\ \vdots \\ z_m \end{bmatrix}}_{z \in \mathbb{R}^m} = \underbrace{\begin{bmatrix} h_{11} & h_{1n} \\ \vdots & \vdots \\ h_{m1} & h_{mn} \end{bmatrix}}_{H \in \mathbb{R}^{m \times n}} \cdot \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{x \in \mathbb{R}^n} + \underbrace{\begin{bmatrix} w_1 \\ \vdots \\ w_m \end{bmatrix}}_{w \in \mathbb{R}^m}$
m Measurements, n States, $w_i \sim \mathcal{N}(0, R)$ – Zero-mean additive gaussian, $x, w_i \forall i$ assumed mutually independent
3.3 Likelihood Estimators
3.3.1 ML-Maximum Likelihood
Choose x to make $\bar{z}$ most likely:
$\underbrace{x \in \mathcal{X}}_{p_x(\bar{x}) \text{ unknown}} \left\{ \underbrace{\bar{x}^{ML} := \operatorname{argmax}_{\bar{x} \in \mathcal{X}} [p_{z x}(\bar{z} \bar{x})]} \right.$
$p_{z x}(\bar{z} \bar{x})$ is var. – technically not a PDF
3.3.2 MAP-Maximum A Posteriori

ML given observation & prior belief about x :
$\underbrace{x \in \mathcal{X}}_{p_x(\bar{x}) \text{ known}} \left\{ \underbrace{\hat{x}^{MAP} := \operatorname{argmax}_{\bar{x} \in \mathcal{X}} [p_{z x}(\bar{z} \bar{x})p_x(\bar{x})]} \right.$
$\hat{x}^{MAP} = \hat{x}^{ML}$ for constant $p_x(\bar{x})$
3.4 General Measurement
$\underbrace{\begin{bmatrix} z_1 \\ \vdots \\ z_m \end{bmatrix}}_{z \in \mathbb{R}^m} = \underbrace{\begin{bmatrix} h_{11} & h_{1n} \\ \vdots & \vdots \\ h_{m1} & h_{mn} \end{bmatrix}}_{H \in \mathbb{R}^{m \times n}} \cdot \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{x \in \mathbb{R}^n} + \underbrace{\begin{bmatrix} w_1 \\ \vdots \\ w_m \end{bmatrix}}_{w \in \mathbb{R}^m}$
m Measurements, n States, $w_i \sim \mathcal{N}(0, R)$ – Zero-mean additive gaussian, $x, w_i \forall i$ assumed mutually independent
3.5 Likelihood Estimators
3.5.1 ML-Maximum Likelihood
Choose x to make $\bar{z}$ most likely:
$\underbrace{x \in \mathcal{X}}_{p_x(\bar{x}) \text{ unknown}} \left\{ \underbrace{\hat{x}^{ML} := \operatorname{argmax}_{\bar{x} \in \mathcal{X}} [p_{z x}(\bar{z} \bar{x})]} \right.$
$p_{z x}(\bar{z} \bar{x})$ is var. – technically not a PDF
3.5.2 MAP-Maximum A Posteriori
ML given observation & prior belief about x :
$\underbrace{x \in \mathcal{X}}_{p_x(\bar{x}) \text{ known}} \left\{ \underbrace{\hat{x}^{MAP} := \operatorname{argmax}_{\bar{x} \in \mathcal{X}} [p_{z x}(\bar{z} \bar{x})p_x(\bar{x})]} \right.$
$\hat{x}^{MAP} = \hat{x}^{ML}$ for constant $p_x(\bar{x})$
3.6 Least Squares Estimators
3.6.1 Linear Least Squares
ML estimate when errors are indep. 0-mean, smae variance, $\mathcal{N}$ -distributed:
$\hat{x}^{LS} := \operatorname{argmin}_x \left[(z - Hx)^\top (z - Hx) \right] = (H^\top H)^{-1} H^\top z$
$\hat{x}^{LS} = \hat{x}^{ML}$ for gaussian noise

3.6.2
Weighted Least Squares

$$\hat{x}^{WLS} := \operatorname{argmin}_x \left[(\bar{z} - H\hat{x})^\top R^{-1} (\bar{z} - H\hat{x}) \right]$$

$$= (H^\top R^{-1} H)^{-1} H^\top R^{-1} \bar{z}$$

Incorporate Prior Knowledge on x:
 $r \sim \mathcal{N}(0, P_x) \Rightarrow x := \hat{x} + r, \mathbb{E}\{x\} = \hat{x}_0, \operatorname{Var}\{x\} = P_x$

Introduce Extended System:

$$\tilde{z} = \begin{bmatrix} \hat{x}_0 \\ z(1:k) \end{bmatrix}, \tilde{H} = \begin{bmatrix} \mathbb{I} \\ H(1:k) \end{bmatrix} \tilde{w} = \begin{bmatrix} -r \\ w(1:k) \end{bmatrix}, \tilde{R} = \begin{bmatrix} P_x & 0 \\ 0 & R(1:k) \end{bmatrix}$$

3.6.3 Recursive Least Squares

Interpretation Update $\hat{x}$ recursively when new $\bar{z}(k)$ arrives
Estimation Error: $e(k) = (\mathbb{I} - KH) e(k-1) - Kw(k)$
Objective Minimize MSE with $P(k) = \operatorname{Var}\{e(k)\}$

$$J(k) := \mathbb{E} \left\{ e^\top(k) e(k) \right\} = \mathbb{E} \{ \operatorname{trace}(P(k)) \}$$

Algorithm

Initialization: $\hat{x}(0) = \hat{x}_0, \quad P(0) = P_x = \operatorname{Var}\{x\}$

Recursion: Observe $\bar{z}(k)$, then update:

$$K(k) = P(k-1)H(k)^\top \left(H(k)P(k-1)H(k)^\top + R(k) \right)^{-1}$$

$$\hat{x}(k) = \hat{x}(k-1) + K(k) (\bar{z}(k) - H(k)\hat{x}(k-1))$$

$$P(k) = (\mathbb{I} - K(k)H(k))P(k-1)(\mathbb{I} - K(k)H(k))^\top + K(k)R(k)K(k)^\top$$

4 Kalman Filter

4.1 Kalman Filter

Exact Solution for Bayes Tracking of LTI Systs. with Gauss. Noise.
Model (Linear Time-Invariant)

$$x(k) = A(k-1)x(k-1) + u(k-1) + v(k-1)$$

$$z(k) = H(k)x(k) + w(k)$$

$x(0), v(\cdot), w(\cdot)$ ind.	$x(0) \sim \mathcal{N}(x_0, P_0)$
$v \sim \mathcal{N}(0, Q(k))$	$w \sim \mathcal{N}(0, R(k))$

Auxilliary Variables

Problem Formulation want $p_{x(k)|z(1:k)} \rightsquigarrow$ bayesian tracking provides this **BUT** we have **CRVs** and don't want integrals
 $\Rightarrow$ exploit linearity, GRV to convert to matrix manipulations
Auxilliary Variables:

Init: $x_m(0) = x(0)$

S1: $x_p(k) = A(k-1)x_m(k-1) + u(k-1) + v(k-1)$

S2: $z_m(k) = H(k)x_p(k) + w(k)$

$$p_{x_m(k)}(\xi) := p_{x_p(k)|z_m(k)}(\xi|\bar{z}(k)) \quad \forall \xi$$

Equivalency x conditioned on $z(1:k-1)/z(1:k)$

$$p_{x_p(k)}(\xi) = p_{x(k)|z(1:k-1)}(\xi|\bar{z}(1:k-1))$$

$$p_{x_m(k)}(\xi) = p_{x(k)|z(1:k)}(\xi|\bar{z}(1:k))$$

GRV x_p and x_m are GRVs for $k = 1, 2, \dots$

$$\hat{x}_p(k) := \mathbb{E}\{x_p(k)\} \qquad P_p(k) := \operatorname{Var}\{x_p(k)\}$$

$$\hat{x}_m(k) := \mathbb{E}\{x_m(k)\} \qquad P_m(k) := \operatorname{Var}\{x_m(k)\}$$

Recursive Update Equations

Initialization $\hat{x}_m(0) = x_0, \quad P_m(0) = P_0$

S1 – Prior Update / Prediction Step

$$\hat{x}_p(k) = A(k-1)\hat{x}_m(k-1) + u(k-1)$$

$$P_p(k) = A(k-1)P_m(k-1)A^\top(k-1) + Q(k-1)$$

S2 – A Posteriori Update / Measurement Update

$$P_m(k) = \left(P_p^{-1}(k) + H^\top(k)R^{-1}(k)H(k) \right)^{-1}$$

$$\hat{x}_m(k) = \hat{x}_p(k) + \underbrace{P_m(k)H^\top(k)R^{-1}(k)}_{\text{correction factor}} \underbrace{(\bar{z}(k) - H(k)\hat{x}_p(k))}_{\text{prediction error}}$$

Alternative Equations – all matrices/vectors at time k

S2 – Kalman Filter Sqiare:

$$K = P_p H^\top \left(H P_p H^\top + R \right)^{-1}$$

$$\hat{x}_m = \hat{x}_p + K (\bar{z} - H\hat{x}_p)$$

$$P_m = \underbrace{(\mathbb{I} - KH) P_p}_{\text{cheaper}} = \underbrace{(\mathbb{I} - KH) P_p (\mathbb{I} - KH)^\top + K R K^\top}_{\text{fewer numerical errors}}$$

Equivalent to RLS $\rightsquigarrow$ among linear, unbiased estimators KF minimizes MSE

KF Remarks

Known Data $A(k), H(k), Q(k), R(k) \quad \forall k$ as well as $P_0 \rightsquigarrow$ KF matrices $P_p(k), P_m(k), K(k)$ can be computed offline

$$\hat{x}_p(k) = A(k-1)\hat{x}_m(k-1) + u(k-1)$$

$$\hat{x}_m(k) = \underbrace{(\mathbb{I} - KH)A(k-1)}_{\hat{A}(k)} \hat{x}_m(k-1)$$

$$+ \underbrace{(\mathbb{I} - KH)}_{\hat{B}(k)} u(k-1) + K\bar{z}(k)$$

KF not guaranteed to be optimal – non-linear estimators may do better

4.1.1 Asymptotic Properties of KF

Model assume **Tl System, stationary Distributions** $(A, H, Q, R$ constants)

Change of Variance

$$P_p(k+1) = \left(AP_p(k)A^\top \right) + Q$$

$$- \left(AP_p(k)H^\top \right) \left(HP_p(k)H^\top + R \right)^{-1} \left(HP_p(k)A^\top \right)$$

Scalar Case $P_p(k+1) = \frac{a^2 r P_p(k)}{h^2 P_p(k) + r} + q =: f(P_p(k))$

Summary *Variance Converges to unique steady-state sol'n provided either $|a| < 1$ or if $|a| \geq 1 \Rightarrow h \neq 0$ and $q > 0$*
If slope for small P_∞ is $> 1 \Rightarrow$ unstable ($|a| > 1$)

4.1.2 Detectability / Stabilizability

Detectability Conditions

Pair (A, H) detectable

$\Leftrightarrow$ For deterministic LTI Sys. $(x(k)=Ax(k-1), z(k)=Hx(k))$
 $\lim_{k \rightarrow \infty} z(k) = 0 \Rightarrow \lim_{k \rightarrow \infty} x(k) = 0 \quad \forall x_0 \in \mathbb{R}^n$

$\Leftrightarrow \begin{bmatrix} A - \lambda \mathbb{I} \\ H \end{bmatrix}$ is full rank for all $\lambda \in \mathbb{C}$ with $|\lambda| \geq 1$ (PBH-Test)

$\Leftrightarrow$ Eigenvalues of $A - LH$ (or $(\mathbb{I} - LH)A$) can be placed within unit circle by suitable choice of $L \in \mathbb{R}^{n \times m}$

Stabilizability Conditions

Pair (A, B) stabilizable

$\Leftrightarrow$ For deterministic LTI Sys. $(x(k)=Ax(k-1)+Bu(k-1))$
 $\exists u(0:k-1)$ s.t. $\lim_{k \rightarrow \infty} x(k) = 0 \quad \forall x_0 \in \mathbb{R}^n$

$\Leftrightarrow \begin{bmatrix} A - \lambda \mathbb{I} & B \end{bmatrix}$ full rank $\forall \lambda \in \mathbb{C}$ with $|\lambda| \geq 1$ (PBH-Test)

$\Leftrightarrow$ Eigenvalues of $A - BK$ (or $(\mathbb{I} - BK)A$) can be placed within unit circle by suitable choice of $K \in \mathbb{R}^{m \times n}$

NOTE Stabilizability is dual of detectability, (A, B) stabilizable

Observability / Reachability

if $(A^\top, B^\top)$ detectable

Observability Matrix $\mathcal{O} = \begin{bmatrix} H \\ HA^{n-1} \end{bmatrix}$

implies detectability

Reachability Matrix $\mathcal{R} = \begin{bmatrix} A & A^{n-1}B \end{bmatrix}$

implies stabilizability

4.1.3 Steady-State KF

DARE – Discrete Algebraic Ricatti Equation (P_∞ is P_p)

$$P_\infty = AP_\infty A^\top + Q - AP_\infty H^\top (HP_\infty H^\top + R)^{-1} HP_\infty A^\top$$

Steady-State KF Gain $K_\infty = P_\infty H^\top (HP_\infty H^\top + R)^{-1}$

Filter Equations Linear Time-Invariant System

$$\hat{x}(k) = \underbrace{(\mathbb{I} - K_\infty H)A}_{\hat{A}} \hat{x}(k-1) + \underbrace{(\mathbb{I} - K_\infty H)}_{\hat{B}} u(k-1) + K_\infty \bar{z}(k)$$

Error dynamics (also holds for non ss-KF) $e(k) = x(k) - x_f(k)$
 $e(k) = (\mathbb{I} - K_\infty H)Ae(k-1) + (\mathbb{I} - K_\infty H)v(k-1) - K_\infty w(k)$
 $\mathbb{E}\{e(k)\} = (\mathbb{I} - K_\infty H)A \mathbb{E}\{e(k-1)\}$

- $(\mathbb{I} - K_\infty H)A$ must be stable (all $|\lambda| < 1$) for e not to diverge

- $\mathbb{E}\{e(k)\} \xrightarrow{k \rightarrow \infty} 0$ iff $(\mathbb{I} - K_\infty H)A$ stable

Theorem for $R > 0, Q \geq 0, G$ s.t. $Q = GG^\top$

$\Leftrightarrow (A, H)$ is detectable, (A, G) is stabilizable

$\Leftrightarrow$ DARE has unique pos. semidef. sol. $P_\infty \geq 0$, resulting $(\mathbb{I} - K_\infty H)A$ is stable and $\lim_{k \rightarrow \infty} P_p(k) = P_\infty \quad \forall P_p(1) \geq 0$

(A,H) Detectable: can observe all unstable modes

(A,G) Stabilizable: noise excites unstable modes

Note if $Q > 0 \Rightarrow (A, G)$ always stabilizable
If $Q = 0$ multiple valid solutions can be possible

4.2	Extended Kalman Filter				
Extension of KF to non-linear systems:					
$x(k) = q_{k-1}(x(k-1), v(k-1))$ $z(k) = h_k(x(k), w(k))$					
<table> <tr> <td>$x(0), v(\cdot), w(\cdot)$ mutually ind.</td><td>$x(0) \sim \mathcal{N}(x_0, P_0)$</td></tr> <tr> <td>$v \sim \mathcal{N}(0, Q(k))$</td><td>$w \sim \mathcal{N}(0, R(k))$</td></tr> </table>	$x(0), v(\cdot), w(\cdot)$ mutually ind.	$x(0) \sim \mathcal{N}(x_0, P_0)$	$v \sim \mathcal{N}(0, Q(k))$	$w \sim \mathcal{N}(0, R(k))$	
$x(0), v(\cdot), w(\cdot)$ mutually ind.	$x(0) \sim \mathcal{N}(x_0, P_0)$				
$v \sim \mathcal{N}(0, Q(k))$	$w \sim \mathcal{N}(0, R(k))$				

EKF Equations

Initialization: $\hat{x}_m(0) = x_0, \quad P_m(0) = P_0$
S1 – Prior Update / Prediction Step

$$\hat{x}_p(k) = q_{k-1}(\hat{x}_m(k-1), 0)$$

$$P_p(k) = A(k-1)P_m(k-1)A^\top(k-1)$$

$$+ L(k-1)Q(k-1)L^\top(k-1)$$

$$A(k-1) = \frac{\partial q_{k-1}(\hat{x}_m(k-1), 0)}{\partial x}, \quad L(k-1) = \frac{\partial q_{k-1}(\hat{x}_m(k-1), 0)}{\partial v}$$

S2 – A Posteriori Update / Measurement Update (at k)

$$K = P_p H^\top \left(H P_p H^\top + M R M^\top \right)^{-1}$$

$$\hat{x}_m = \hat{x}_p + K(\bar{z} - h_k(\hat{x}_p(k), 0))$$

$$P_m = (\mathbb{I} - K(k)H(k)) P_p(k)$$

$$H(k) := \frac{\partial h_k(\hat{x}_p(k), 0)}{\partial x}, \quad M(k) := \frac{\partial h_k(\hat{x}_p(k), 0)}{\partial w}$$

Intuition
Process Update – predict the mean state estimate fwd using NL process model and update the var. according to lin. eqns
Measurement Update – correct for mismatch between $\bar{z}(k)$ and prediction $h_k(\hat{x}_p(k), 0)$, and correct var. according to lin. eqn
Caution

- Process & measurement noise **assumed** 0-mean
- A, L, H, M lin. about current $\hat{x}_m \rightsquigarrow$ **cannot** compute offline
- Prior update only accurate if $\mathbb{E}\{\cdot\}$ & $q_{k-1}(\cdot)$ commute
- EKF Variables DO NOT capture true conditional mean & var.**

4.2.1 Hybrid EKF

System Process – non-lin, CT, **Measurement** – non-lin, DT

$$\dot{x}(k) = q(x(t), v(t), t) \quad x[k] := x(kT)$$

$$z[k] = h_k(x[k], w[k]) \quad \mathbb{E}\{w[k]\} = 0, \text{Var}\{w[k]\} = R$$
CT White Noise $\delta(t)$ = Dirac pulse

$$\mathbb{E}\{v(t)\} = 0 \text{ and } \mathbb{E}\{v(t)v^\top(t+\tau)\} = \mathcal{Q}_c\delta(\tau)$$
True CT White noise does not exist $\rightarrow$ Would require infinite power

Hybrid EKF Equations

Initialization $\hat{x}_m(0) = x_0, \quad P_m(0) = P_0$
S1 – Prior Update / Prediction Step (Solve ODEs)
Solve, for $(k-1)T \leq t \leq kT$ and $\hat{x}((k-1)T) = \hat{x}_m[k-1]$

$$\dot{\hat{x}}(t) = q(\hat{x}(t), 0, t) \quad \Rightarrow \hat{x}_p[k] := \hat{x}(kT)$$
Solve, for $(k-1)T \leq t \leq kT$ and $P((k-1)T) = P_m[k-1]$

$$\dot{P}(t) = AP + PA^\top + LQ_cL^\top \quad \Rightarrow P_p[k] := P(kT)$$
with

$$A(t) := \frac{\partial q}{\partial x}(\hat{x}(t), 0, t), \quad L(t) := \frac{\partial q}{\partial v}(\hat{x}(t), 0, t)$$
S2 – A Posteriori Update / Measurement Update
Measurement model still DT – S2 identical to standard EKF
Caution

- Process & measurement noise **assumed** 0-mean
- ODEs typically solved numerically: Accuracy $\uparrow$ = Comp. Cost $\uparrow$

5 Particle Filter

5.1 Monte Carlo Sampling

Samples N DRVs y^n of samples of $y \in \mathcal{Y} = \{1, \dots, \bar{Y}\} \sim p_y$
Helper Variables

$$s_i^n = \delta(i - y^n) = \begin{cases} 1 & \text{if } y^n = i \\ 0 & \text{otherwise} \end{cases} \quad \begin{matrix} i = 1, \dots, \bar{Y} \\ n = 1, \dots, N \end{matrix}$$

Average – s_i converges to $p_y(i)$ by Law of Large Numbers

$$\frac{1}{N} \sum_{n=1}^N s_i^n \xrightarrow{N \rightarrow \infty} \mathbb{E}\{s_i^n\} = p_y(i) \quad p_y(i) \approx \frac{1}{N} \sum_{n=1}^N \bar{s}_i^n$$

Approximation
For $y \in \mathcal{Y}$, it holds $p_y(i) \approx \frac{1}{N} \sum_{n=1}^N \delta(i - \bar{y}^n), \quad i \in \mathcal{Y}$
For $x = g(y)$, with $x \in \mathcal{X} := g(\mathcal{Y})$, it holds

$$p_x(j) \approx \frac{1}{N} \sum_{n=1}^N \delta(j - g(\bar{y}^n)), \quad j \in g(\mathcal{Y})$$

Monte Carlo Sampling CRV

Helper Variables

$$s_a^n = \int_a^{a+\Delta y} \delta(i - y^n) d\xi = \begin{cases} 1 & \text{if } a \leq y^n < a + \Delta y \\ 0 & \text{otherwise} \end{cases}$$

Average – s_a converges to $p_y(\xi)$ by Law of Large Numbers

$$\int_a^{a+\Delta y} \frac{1}{N} \sum_{n=1}^N \delta(\xi - y^n) d\xi \xrightarrow{N \rightarrow \infty} \int_a^{a+\Delta y} p_y(\xi) d\xi$$
Approximation

For $y \in \mathcal{Y}$, it holds $p_y(\xi) \approx \frac{1}{N} \sum_{n=1}^N \delta(\xi - \bar{y}^n) \forall \xi$

For $x = g(y)$, it holds $p_x(\xi) \approx \frac{1}{N} \sum_{n=1}^N \delta(\xi - g(\bar{y}^n)) \forall \xi$

5.2 Modeling System Non-linear, discrete-time:

$$x(k) = q_{k-1}(x(k-1), v(k-1))$$

$$z(k) = h_k(x(k), w(k))$$

$x(0), \{v\}, \{w\}$ mutually indep. can be D- or CRVs, known PDFs
Objective
Bayesian State Estimator with Monte Carlo Sampling

Init:	$x_m(0)$	$:= x(0)$
S1:	$x_p(k)$	$:= q_{k-1}(x_m(k-1), v(k-1))$
S2:	$z_m(k)$	$:= h_k(x_p(k), w(k))$
	$p_{x_m(k)}(\xi)$	$:= p_{x_p(k) z_m(k)}(\xi \bar{z}(k)) \quad \forall \xi$

Similar proof for $p_{x_p(k)}(\xi), p_{x_m(k)}(\xi) \forall \xi$ as for KF aux. var.

5.3 Particle Filter - Equations
Initialization Draw N samples $\{\bar{x}_m^n(0)\}$ from $p_{x(0)}$
S1 – Prior Update / Prediction Step
Apply process equation to the particles $\{\bar{x}_m^n(k-1)\}$

$$\bar{x}_p^n(k) := q_{k-1}(\bar{x}_m^n(k-1), \bar{v}^n(k-1)), \quad n = 1, \dots, N$$
Requires N noise samples from $p_{v(k-1)}$
S2 – A Posteriori Update / Measurement Update
Scale each particle by measurement likelihood:

$$\beta_n = \alpha p_{z(k)|x(k)}(\bar{z}(k)|\bar{x}_p^n(k)), \quad n = 1, \dots, N$$

with α normalization s.t $\sum_{n=1}^N \beta_n = 1$

$$\alpha = \left(\sum_{n=1}^N p_{z_m(k)|x_p(k)}(\bar{z}(k)|\bar{x}_p^n(k)) \right)^{-1}$$

Resample to get N posterior particles $\{\bar{x}_m^n(k)\}$, w/ eq. weights

Repeat N times:
- Select random number r uniformly on $(0, 1)$ - Pick particle $\bar{n}$ s.t $\sum_{n=1}^{\bar{n}-1} \beta_n < r \leq \sum_{n=1}^{\bar{n}} \beta_n$

New particles all have equal weight

$$p_{x_m(k)}(\xi) \approx \frac{1}{N} \sum_{n=1}^N \delta(\xi - \bar{x}_m^n(k)), \quad \forall \xi$$

Intuition
Process Update – Propagate N particles through system dynamics. Provided N large, $\{x_m^n(k-1)\} \approx p_{x_m(k-1)} \rightsquigarrow \{\bar{x}_p^n(k)\} \approx p_{x_p(k)}$
Measurement Update – treat each particle separately. At points of high prior, we have many particles. Posterior has same particles, but scaled by measurement likelihood.

5.4 Roughening Sample Impoverishment Lumpiness

– when we resample, only retain subset of particles. For finite N this is problem.
Roughening
Perturb particles **after** resampling

$$\bar{x}_m^n(k) \leftarrow \bar{x}_m^n(k) + \Delta x^n(k)$$
 $\Delta x^n(k)$ drawn from 0-mean, finite-var. distribution
Example Method for Δx^n : σ_i stand. dev. of $\Delta x_i^n(k)$

$$\sigma_i = K E_i N^{-\frac{1}{d}}$$

K: tuning parameter, typically $K \ll 1$
d: dimension of state space

$$E_i: \text{max inter-sample variability } \max_{n_1, n_2} |\bar{x}_{m,i}^{n_1} - \bar{x}_{m,i}^{n_2}|$$

$$N^{-\frac{1}{d}}: \text{node spacing on grid}$$

6 Observer Based Control